The Substitution Rule (5.5)

$$\int e^{x^2} dx$$

$$\int x e^{x^2} dx$$

Recall the chain rule:

$$(f(g(x))' = f'(g(x))\,g'(x).$$

In other words, $f(g(x))$ is an antiderivative of $f'(g(x))\,g'(x)$. Thus,

$$\int f'(g(x))g'(x)\,dx = f(g(x)) + C$$

In practice, it is often hard to recognize functions of the form $f'(g(x))\,g'(x)$.

Substitution rule:

Let $u = g(x)$. Then

$$du = g'(x)dx$$

and we can replace any instance of $g'(x)dx$ in an integral with du.

$$\int x^2 \sin(x^3+1)\,dx$$

$$\int \frac{x}{\sqrt{1+x^2}}\,dx$$

$$\int \frac{x^3}{\sqrt{1+x^2}}\,dx$$

$$\int \frac{\ln x}{x} dx$$

$$\int_0^1 x\sqrt{2x^2+1}\,dx$$

$$\int_0^2 x\sqrt{3-x}\,dx$$

$$\int_{3\pi/4}^{\pi} \tan x \, dx$$

$$\int_{-1}^{1} x\sqrt{1-x^2}\,dx$$

$$\int_{-1}^{1} \sqrt{1-x^2}\, dx$$